

COVID-19 Data Analysis — Final Project Report

1. Introduction

source of data: Our World in Data (OWID) COVID-19 Dataset

<https://ourworldindata.org/coronavirus>

analyze the tendency by the COVID data of global new cases, vaccination vs death rate over time, total deaths and cases by continent.

main goal of this project:

- use pandas to clean the Data (OWID) COVID
- use matplotlib to show the plots of tendency and rate
- use Plotly express to create a interactive plot
- make a conclusion based on the results I got from my analysis

2. Data cleaning part

The following steps were applied to prepare the OWID COVID-19 dataset for analysis and visualization:

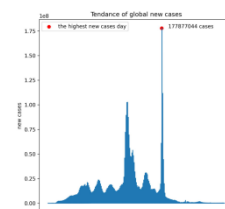
- Convert the date column to a proper "datetime" format, ensuring consistent time-series operations.
- Fill missing numerical values using column means, reducing bias caused by incomplete data and improving the stability of downstream calculations.
- Standardize country names (e.g., US, U.S.A., USA → United States), ensuring that all records referring to the same country are grouped correctly.
- Remove invalid rows or entries with missing dates to avoid errors in time-series visualization and grouping.
- Compute 7-day rolling averages for key indicators such as new cases and new deaths, providing a smoother and more interpretable trend for plotting and mapping.
- Save the cleaned dataset to the file: data_clean/owid-covid-data-clean.csv, which serves as the unified data source for all subsequent visualizations.

These preprocessing steps ensure internal consistency, improve analytical reliability, and provide a clean, standardized foundation for all static and interactive plots.

3. Visualization by matplotlib

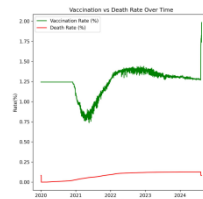
Tendency of new global cases

-with the plot, we can see that since 2020, at least 6 outbreaks have occurred, with 2 serious ones at the beginning of 2022 and 2023 respectively. And since the middle of 2023, the cases have plummeted.



Vaccination vs Deaths Rate over time

- at the far right of the plot, it is observed that with the vaccination rate going up significantly, the death rate went down sharply. But in the central area of the plot, we cannot see a clear relationship between the vaccination and death rate.

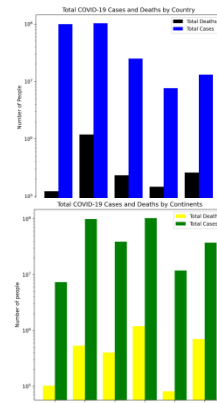


Total COVID-19 Cases and Deaths by country

- we can tell that although USA and China have got almost the same number of cases, the deaths in USA are far more than those in China and that Iran has the lowest cases among all the five countries listed here.

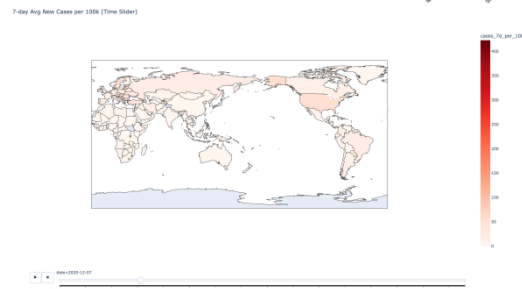
Total COVID-19 Cases and Deaths by continent

- we can see that clearly that the rate between deaths and cases in Oceania is the lowest, and in South America is the highest, perhaps we can connect it to the economy and the medical environment in those continents, that will directly affect the death rate.



4. Interactive Visualisation by Plotly

By using the Plotly Express, we can see all the countries of the number of people in every 7 days per 100k, we can also choose any specific date from 2020 to 2024, and we can clearly see the global distribution of pandemic outbreaks.

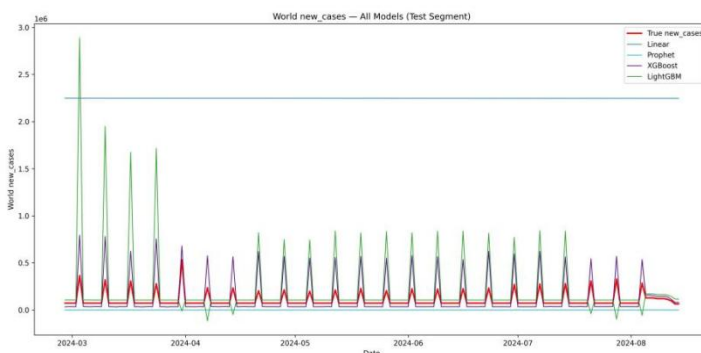


And more precisely:

- During certain periods, Europe and the Americas appear in the darkest colors, indicating rapid transmission and high infection intensity.
- As time progresses, the color gradually fades, suggesting a general easing of the pandemic or a reduction in newly reported cases.
- By normalizing case counts per 100,000 population, the map avoids biases caused by large population countries, allowing for a more meaningful comparison across regions.

5. Forecasting

By using the Sklearn, Prophet, XGboost and Lightgbm libraries, I first cut out the last 10% of the data, then attempted to forecast the new cases in the world, and finally calculated RMSE and MAE to evaluate which model performed the best.



The red curve represents the true values from the dataset.

The blue horizontal line at the top corresponds to the Linear Regression model. In this case, I think the Linear Regression serves more as a baseline rather than a realistic predictor, because the new cases are fluctuating sharply, using a straight line to simulate future trends is clearly inadequate.

The green curve shows predictions from the Lightgbm model, while the purple curve comes from Xgboost model. The Lightgbm model is very similar to the Xgboost model, because these two

models both need many different data related to the data of new cases, but in my experiment, I find that the Xgboost performs better. I am not sure why this happens. It may be due to the mechanism behind.

The light blue line at the bottom represents the Prophet model. As I understood, Prophet is a time-based forecasting model. While its shape resembles that of the Linear Regression with time feature, it is more precise because it analyses not only the daily trends but also the week, the season and the year patterns.

From the table on the right, there are two different ways of calculating the error: RMSE and MAE. In terms of RMSE, both Prophet and XGboost perform well, while Lightgbm does not perform as well. In terms of MAE, however, XGBoost is much better than the other models, whereas Prophet and Lightgbm perform equally well.

	RMSE	MAE
Linear	2.147196e+06	2.145965e+06
Prophet	1.245631e+05	1.011664e+05
XGBoost	1.312973e+05	7.681464e+04
LightGBM	3.306636e+05	1.297872e+05

6. Key findings

In early 2022 and 2023, we can see that two sudden explosions of the COVID pandemic, caused by new virus variants, created a major public health crisis.

And we can see that there is a negative correlation between vaccination rates and death rates, which indicates that vaccinations are effective in reducing deaths among high-risk patients.

From the data of each continent, we can conclude that the death rate is closely related to the local economy.

From the interactive figure using Plotly, we can see that as time passes by, COVID, which originally started from Asia, spread to Europe and America afterward.)

From this kind of data, I have found that the XGBoost model performs best in forecasting the causes, and that it can predict future outcomes based on various related data.

7. Conclusion

It's a small project where we can put Numpy, Pandas, Mathplotlib, Plotly, Sklearn, Xgboost, Prophet, Lightgbm and calculate the error into real practice, and we learn that we need to clean the data before visualising that data. And this project enables a multi-dimensional analysis of the COVID-19 data, but it has its limitations.